

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination – 2022

Course: B. Tech.

Branch :Electronics & Telecommunication Engineering

Semester :VII Subject Code & Name: BTETPE703A Embedded System Design

Max Marks: 60

Date: 01/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Briefly explain three different types of RTOS.	Level 4	6
B) Explain different characteristics of an Embedded System.	CO1	6
C) Draw and explain basic structure of an Embedded System.	CO1	6
Q.2 Solve Any Two of the following.		12
A) Distinguish between Von-Neumann and Harvard Architecture.	Level 3	6
B) Explain in brief about following non-functional requirements needs to be specified before designing an Embedded System. Performance, Cost, & Power Consumption.	CO3	6
C) Explain in brief about Erasable Programmable Read Only Memory(EPROM)	CO2	6
Q. 3 Solve Any Two of the following.		12
A) State different characteristics of Digital Signal Processors.	Level 4	6
B) Explain in brief about I2C bus interfacing with microprocessor / controller.	CO2	6
C) Distinguish between RISC and CISC systems.	Level 1	6
Q.4 Solve Any Two of the following.		12
A) Distinguish between Hard Real Time Systems and Soft Real Time System.	Level 2	6
B) Explain in brief Task/Process Scheduling, Task/Process Synchronization & Error/Exception handling.	Level 2	6
C) Explain in brief about interrupt handling in RTOS.	CO4	6
Q. 5 Solve Any Two of the following.		12
A) Explain in brief Co-operative Multitasking, Preemptive Multitasking & Non-preemptive Multitasking.	CO4	6
B) Distinguish between Thread & Process.	Level 2	6
C) Explain Rate Monotonic (RM) Scheduling Algorithm with suitable example.	Level 2	6

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